

Mistake of Proofs & Mathematical Induction



- Direct Proofs
- Indirect Proofs
 - Proof of the Contrapositive
 - Proof by Contradiction

Theorems that are Bi conditional
Statements

To prove a theorem that is a bi conditional statement, that is, a statement of the form $p \leftrightarrow q$, we show that $p \rightarrow q$ and $q \rightarrow p$ are both true.

- **Example:** Prove the theorem: “If n is an integer, then n is odd if and only if n^2 is odd.”

- Direct Proofs
- Indirect Proofs
 - Proof of the Contrapositive
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Theorems that are Bi conditional
Statements

Solution: We have already shown (previous slides) that both $p \rightarrow q$ and $q \rightarrow p$. Therefore we can conclude $p \leftrightarrow q$.

- Mistake of proof

What is wrong with this?

Proof" that $1 = 2$

Let $a = b$

premise

$$a^2 = ab$$

Multiply both sides
by a

$$a^2 - b^2 = ab - b^2$$

Subtract b^2 both sides

$$(a + b)(a - b) = b(a - b)$$

Algebra on

$$(a + b) = b$$

Divide both side by $(a - b)$

$$2 = 1$$

Divide both side by b

Mathematical Induction

Principle of Mathematical Induction: To prove that $P(n)$ is true for all positive integers n , we complete these steps:

- ***Basis Step:*** Show that $P(1)$ is true.
- ***Inductive Step:*** Show that $P(k) \rightarrow P(k + 1)$ is true for all positive integers k .

To complete the inductive step, assuming the *inductive hypothesis* that $P(k)$ holds for an arbitrary integer k , show that $P(k + 1)$ must be true.

Prove that $1+2+3+\dots+n = \frac{n(n+1)}{2}$

Basis step: $p(1)$ is true since $\frac{1(1+1)}{2} = 1$

Inductive Step: Assume that the statement true for $p(k)$:

That is $p(k)=1+2+3+\dots+k = \frac{k(k+1)}{2}$

We have to show that $p(k+1)$ is true.

Under the assumption of inductive steps, we have

$$p(k)=1+2+3+....+k=\frac{k(k+1)}{2}$$

$$\Rightarrow p(k+1)=1+2+3+....+k+(k+1)=\frac{k(k+1)}{2}+(k+1)$$

$$\Rightarrow \frac{k(k+1)+2(k+1)}{2}$$

$$\Rightarrow \frac{(k+1)(k+2)}{2}$$

$\Rightarrow p(k+1)$ statement is true

$\therefore$ the statement is true

That is $1+2+3+\dots+n = \frac{n(n+1)}{2}$
true for all natural number.

Example: Use mathematical induction to prove that $n < 2^n$ for all positive integers n .

Solution: Let $P(n)$ be the proposition that $n < 2^n$.

– **BASIS STEP:** $P(1)$ is true since $1 < 2^1 = 2$.

– **INDUCTIVE STEP:** Assume $P(k)$ holds, i.e., $k < 2^k$, for an arbitrary positive integer k .

We have to show that $p(k+1)$ is true.

Must show that $P(k + 1)$ holds. Since by the inductive hypothesis, $k < 2^k$, it follows that:

$$k + 1 < 2^k + 1 \leq 2^k + 2^k = 2 \cdot 2^k = 2^{k+1}$$

Therefore $n < 2^n$ holds for all positive integers n .

Example: Use mathematical induction to prove that $2^n < n!$, for every integer $n \geq 4$.

Solution: Let $P(n)$ be the proposition that $2^n < n!$.

– **BASIS STEP:** $P(4)$ is true since $2^4 = 16 < 4! = 24$.

– **INDUCTIVE STEP:** Assume $P(k)$ holds, i.e., $2^k < k!$ for an arbitrary integer $k \geq 4$.

To show that $P(k + 1)$ holds:

$$2^{k+1} = 2 \cdot 2^k$$

$$< 2 \cdot k! \quad (\text{by the inductive hypothesis})$$

$$< (k + 1)k!$$

$$= (k + 1)!$$

$$= (k + 1)!$$

Therefore, $2^n < n!$ holds, for every integer $n \geq 4$.

Note that here the basis step is $P(4)$, since $P(0)$, $P(1)$, $P(2)$, and $P(3)$ are all false.

Example: Use mathematical induction to prove that $n^3 - n$ is divisible by 3, for every positive integer n .

Solution: Let $P(n)$ be the proposition that $n^3 - n$ is divisible by 3.

– **BASIS STEP:** $P(1)$ is true since $1^3 - 1 = 0$, which is divisible by 3.

– **INDUCTIVE STEP:** Assume $P(k)$ holds, i.e., $k^3 - k$ is divisible by 3, for an arbitrary positive integer k .

To show that $P(k + 1)$ holds:

$$(k + 1)^3 - (k + 1) = (k^3 + 3k^2 + 3k + 1) - (k + 1)$$

$$= (k^3 - k) + 3(k^2 + k)$$

By the inductive hypothesis, the first term $(k^3 - k)$ is divisible by 3 and the second term is divisible by 3 since it is an integer multiplied by 3.

$\Rightarrow (k + 1)^3 - (k + 1)$ is divisible by 3.

Therefore, $n^3 - n$ is divisible by 3, for every integer positive integer n .

